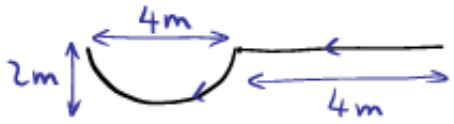
P1 May 2010 TZ2

1.

1. Given that $I = 2.00A \pm 0.02A$ the absolute uncertainty is $\pm 0.02A$ (No trick)

The percentage uncertainty can be calculated from $\frac{0.02}{2.00} \times 100\% = 1\%$ (A)

2.

2. 

The magnitude of displacement is the direct distance from P→Q

Since the curved part is a semicircle the distance across it is 2× radius
Total distance = 8m

(D)

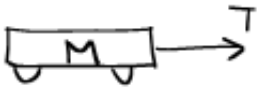
3.

3. Applying Newton's 2nd Law to the hanging mass



$\rightarrow W - T = ma$

And to the cart



$T = Ma$

Substituting for T $\rightarrow$ $W - Ma = ma$

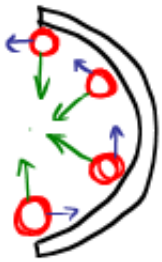
$W = (M+m)a \Rightarrow a = \frac{W}{M+m}$

but $m = \frac{W}{g} \Rightarrow a = \frac{W}{M+\frac{W}{g}}$ (B)

A simpler way is to use the units a → force/mass
B is the only option that has force/mass

4.

4.



Since motion is circular the force is directed towards the centre

The average force will be in the direction of change in momentum



change of momt = final − initial



(B)

5.

5. Power = $\frac{\text{Work done}}{\text{time taken}}$

Work done lifting mass m height h = mgh

Power = $\frac{mgh}{t}$ but $\frac{m}{t}$ = mass per unit time, R

so P = Rgh (B)

or use units; Rgh is the only one with units of Power

6.

6. This question led to lengthy discussion. Clearly the box gains KE but what about the molecules of gas?

First thought is that although the gas molecules gain KE as a whole they don't gain KE individually so U and T do not increase.

However when accelerated the mols will be hit by the wall of the container so will gain KE. This gain is lost at the other end which is moving away.

I made an animation to show this. The gas gets hot but then cools

⇒ answer A.

May12 0123.wmv

7.

7. Applying the equation $Q = mc\Delta T$

$$\frac{\text{thermal energy to P}}{\text{thermal energy to Q}} = \frac{m_P c_P \Delta T}{m_Q c_Q \Delta T}$$

Temp. change is the same for each so

$$= \frac{m_P c_P}{m_Q c_Q}$$ (B)

8.

8. The gas is ideal so $PV = nRT$

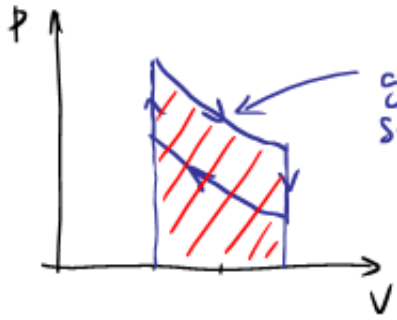
If temp. const. then $PV \propto n$

so doubling n will double PV gives $2p_oV_o$.

(D)

9.

9.



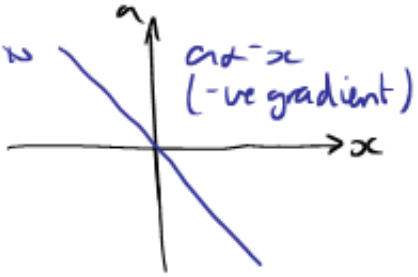
gas expands here
so WD = Area under this line
= area I + II

(C)

10.

10. From the definition of SHM $a \propto -x$

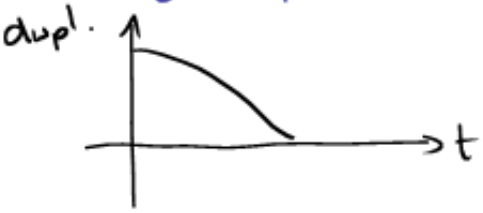
so the graph of a vs x is



(B)

11.

11. A car shock absorbers (dampers) should stop the car going past the equilibrium position

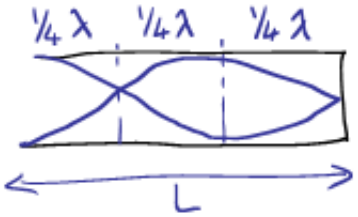


This is critical damping

(D)

12.

12.



second harmonic when

$L = \frac{3}{4}\lambda$

$\Rightarrow \lambda = \frac{4L}{3}$

13.

13.



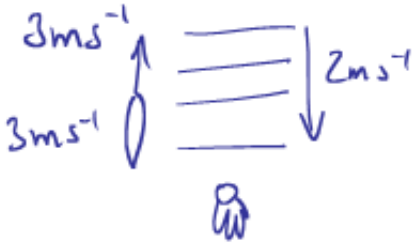
The frequency is higher for the boat so it must be travelling towards the waves (away from shore)

$v = f\lambda \Rightarrow \lambda = \frac{2}{0.2} = 10m$

For boat $f = 0.5Hz \Rightarrow v = 0.5 \times 10 = 5ms^{-1}$
so relative velocity between boat + waves $= 5ms^{-1}$

Velocity relative to shore $= 3ms^{-1}$

(A)



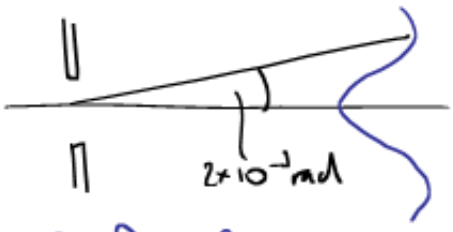
14.

14. An LCD rotates the plane of polarisation (C)

It also "changes colour" when the segment is to be displayed but this is not the best answer.

15.

15.



$d\sin\theta = n\lambda$

If θ small in rads

$\theta = \lambda/d$

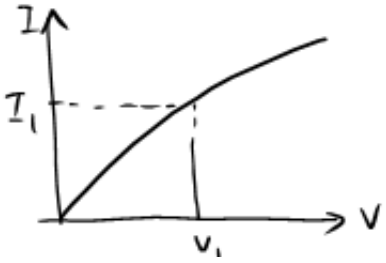
so $\theta \propto \lambda$

$\frac{\theta_1}{\theta_2} = \frac{\lambda_1}{\lambda_2} \quad \frac{2\times10^{-3}}{\theta_2} = \frac{600}{400} \Rightarrow \theta_2 = 1.3\times10^{-3}rad$

(C)

16.

16.



Resistance is the ratio $\frac{V_1}{I_1}$

(A)

17.

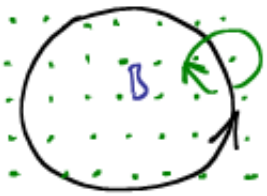
17.



so $R_T = 5\Omega$ (B)

18.

18.



If B decreases then EMF will be induced (faraday)

The direction of current will oppose the change (lenz) so will maintain the B field

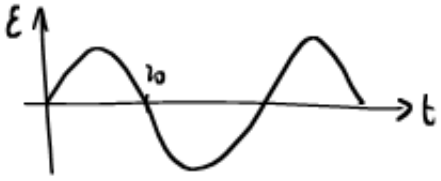
⇒ Anticlockwise

As rate of change of B is constant then I will be constant.

(D)

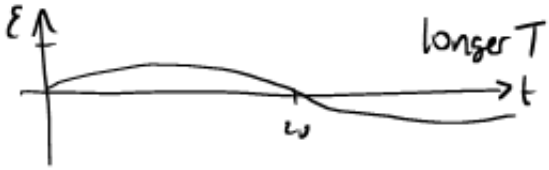
19.

19.



If frequency is ½ the Time period will be 2×

This means the rate of change of B will also be less ⇒ induced ε is less

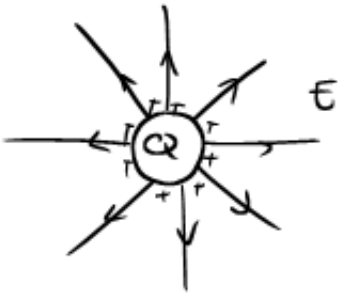


longer T, lower ε

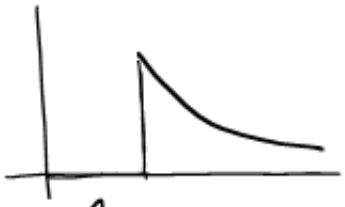
(A)

20.

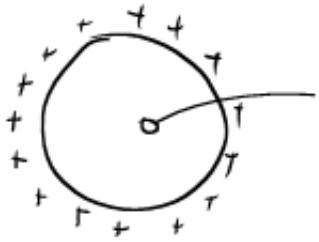
20.



$\mathcal{E} = \frac{kQ}{r^2}$



inside sphere $\mathcal{E} = 0$

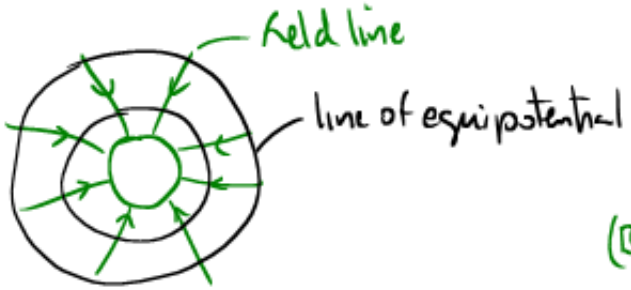


Forces cancel

(D)

21.

21. gravitational field lines are always ⊥ to lines of equipotential.



(B)

22.

22. The equation for $V = \frac{GM}{r}$ so $V \propto 1/r$

Since the force is always attractive V is always -ve

(B)

23.

23. When an astronaut is in orbit the g field is not zero, if it was what would give the centripetal force?

The reason the astronaut feels weightless is because he and his spaceship are in free fall. In other words they have the same acceleration.

(B)

24.

24. α decay results in the proton number reducing by 2

$${}^{A}_{Z}X \rightarrow {}^{A-4}_{Z-2}Y + {}^{4}_{2}\alpha$$

β^- decay results in the proton number increasing by 1

$${}^{A}_{Z}X \rightarrow {}^{A}_{Z+1}Y + \beta \qquad (C)$$

So α followed by $2\beta^-$ will result in the same proton no.

25.

25. If the mass defect = 0.1u

then $\Delta E = 0.1 \times 931.5$ MeV

$= 93$ MeV (A)

B is the wrong unit

26.

26. As α particles pass through air they ionise air atoms. The length of path depends upon the number of ions created which is related to the KE of the α and the density of air.

But II and III only is not an option

. So I assume that the charge is an important factor in the number of ion pairs produced

(D)

27.

27. Photo electric emission occurs when an electron absorbs a photon of light.

If the photon energy is enough the e^- can gain enough energy to leave the surface.

Since $E = hf$ this only happens if f is high.

Which is equivalent to λ being small ($v = f\lambda$) (D)

28.

28.

If electron accelerated $Ve = \frac{1}{2}mv^2$

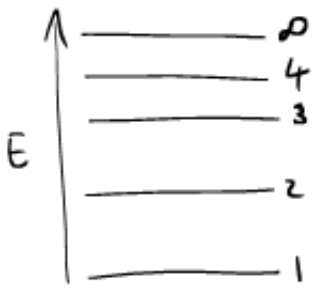
$\Rightarrow v = \sqrt{\frac{2Ve}{m}}$ $p = m\sqrt{\frac{2Ve}{m}}$

de Broglie wavelength $\lambda = \frac{h}{p} = h/m\sqrt{\frac{2Ve}{m}}$

$\Rightarrow \lambda \propto 1/\sqrt{V}$ (D)

29.

29.



$E = \frac{hc}{\lambda} \Rightarrow$ short λ is high E

Largest transition is from 4 → 1 short λ

smallest transition from 4 → 3 long λ

(A)

30.

30. In Schrödingers model the electron were represented by a wave function. This is not quite the same thing as saying that the electrons are waves.

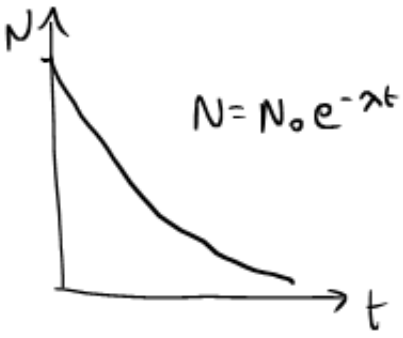
(D)

31.

31. The fact that α's have discrete energies leads us to believe that the nucleus has E levels rather like line spectra providing evidence for electron E levels.

(A)

32.

32. 

this is an exponential curve so the slope (dN/dt) is proportional to N.

(C)

33.

33.

Renewable	Non Renewable
Oil	gas
Wind	coal
Waves	nuclear
geothermal	oil
biofuels	

(C)

34.

34. The gas turbine use first the 'exploding gas' then the heat from the gas to turn turbines. This makes it more efficient.
This is the same as saying more energy/unit mass and less CO_2 per unit energy.

(D)

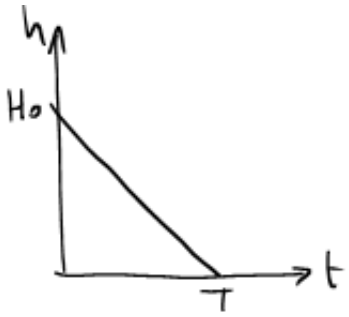
35.

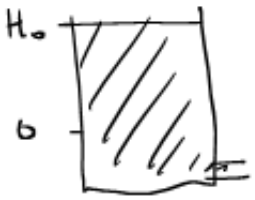
35. The critical mass is the amount of fuel needed to sustain a constant fission rate. More than this and the rate accelerates.

(A)

36.

36.





Power = $\frac{mgh}{T}$

= $\frac{m}{T} \cdot gh$

gradient ↗ ↑ H_0

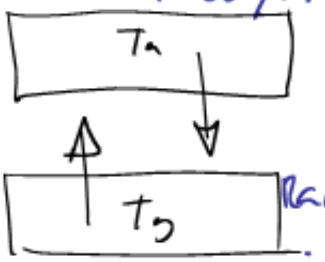
(A)

37.

37. Power radiated/m^2 $= \sigma T^4$ for a black body

for a body with emmisivity ε

Power/m^2 $= \varepsilon \sigma T^4$



for ground $P/m^2 = \sigma T_s^4$
for atmosph. $P/m^2 = \varepsilon \sigma T_a^4$
Area same for each and
Radiation from ground = radiation from atmosphere

so $\cancel{\sigma} T_s^4 = \varepsilon \cancel{\sigma} T_a^4$

$\frac{T_s^4}{T_a^4} = \varepsilon \Rightarrow \frac{T_s}{T_4} = \varepsilon^{1/4}$ (C)

38.

38.



for destructive interference path diff $= \lambda/2$

path diff. $= 2d = \lambda/2$

$d = \lambda/4$ i.e. $d \propto \lambda$

so $\frac{\text{pit height CD}}{\text{pit height DVD}} = \frac{\lambda_{CD}}{\lambda_{DVD}} = \frac{800}{400} = 2$

(B)

39.

39.

N photons → N electrons

⇒ charge $= Ne = Q$

Pixel has capacitance $C = \frac{Q}{V} \Rightarrow V = \frac{Q}{C}$

$V = \frac{Ne}{C}$ (C)

Also can be deduced from units $V = \frac{Q}{C}$

40.

40.

Linear magnification = $\frac{\text{image size}}{\text{object size}}$

for resolution point must be separated by a pixel



so image size = $2l$

object size = D

$M = \frac{2l}{D}$ $\frac{MD}{l} = 2$ (C)

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